



AI Infrastructure Architecture Evolution

2015-2026

Neoverse · CSS · AGI CPU · CMN

Ethos · Mali · Partner Silicon · Systems · Software

A generation-by-generation synthesis of conference disclosures and primary technical sources.

Evidence cutoff: 28 September 2026.

Complete English edition, paired with a full Simplified Chinese report.

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01 Scope and executive synthesis

Families and weights: deep on Neoverse, CSS and AGI CPU; full on mesh/system IP; proportional on licensee silicon, GPU/NPU IP and software.

Start: 2015, with Cortex-A server designs as the baseline.

Cutoff: 28 September 2026; includes Hot Chips 2026 and the September CSS N4 announcement.

Focus: core evolution, coherent memory systems, integration boundaries and the CPU role in AI infrastructure.

Exclusions: exhaustive licensee SKUs and mobile generations; no Arm-branded datacenter training GPU or merchant AI NIC is established in the reviewed portfolio.

Arm's infrastructure evolution has three overlapping layers. Neoverse separates server-oriented core development from a mobile-led product story. Compute Subsystems move validation of the CPU, mesh, memory interfaces and platform software upstream into Arm. AGI CPU adds Arm-designed merchant silicon in 2026. The last transition changes the competitive boundary: Arm can now supply both the building blocks for a custom server and a finished CPU for organizations without a custom-silicon program.

The important progression is not simply more cores. N1 establishes a scalable cloud core; V1 adds substantial vector capacity; N2 emphasizes density and Armv9; V2 widens the scalar machine and improves memory access; V3 and CSS bring confidential computing and more complete integration. CSS N4 then expands configuration and I/O headroom. The realized outcome still depends on the licensee's memory channels, cache, physical design and software. A Neoverse name is therefore insufficient to predict socket performance.

02 Units, ownership and availability

IP availability is not silicon general availability. A core does not have a universal foundry node, package power, socket count or DDR bandwidth. Those properties belong to an implementation. Tables retain the standard cross-vendor columns and use n/a for a field that does not apply, and n/d when reviewed sources do not disclose it. Cloud vCPU limits are not automatically physical core counts. Historical shipping does not imply current orderability.

Memory bandwidth is decimal payload GB/s: channels × transfers per second × eight bytes for a conventional 64-bit DDR channel. Per-core bandwidth is a derived equal-share ratio, not a reservation. Link figures are per direction unless explicitly labelled otherwise. ISA vector length, number of execution pipelines and aggregate bytes processed per cycle are different quantities. A two-pipeline 256-bit SVE core must not be compared with a four-pipeline 128-bit core by vector length alone. Confidential-computing capability likewise requires system and software enablement, not just an ISA label.

03 Neoverse and CSS generation atlas

CPU IP comparison: implementation fields remain separate · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	GA	Status	Core march	Cores / threads
Neoverse N1 / Ares	2019 IP disclosure	Licensable IP	N1	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse E1	2019 IP disclosure	Licensable IP	E1	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse V1 / Zeus	2020-2021 IP disclosure	Licensable IP	V1	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse N2 / Perseus	2020-2021 IP disclosure	Licensable IP	N2	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse V2 / Demeter	2022-2023 IP disclosure	Licensable IP	V2	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse V3	2024 IP disclosure	Licensable IP	V3	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
Neoverse N3	2024 IP disclosure	Licensable IP	N3	Configurable; E1 SMT2; N4 threads n/d; others single-threaded
CSS N4	2026-09 IP disclosure	Licensable IP	N4	Configurable; E1 SMT2; N4 threads n/d; others single-threaded

CPU IP comparison: implementation fields remain separate - 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
Neoverse N1 / Ares	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	512 KB / 1 MB	System configuration
Neoverse E1	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	n/d	System configuration
Neoverse V1 / Zeus	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	512 KB / 1 MB	System configuration
Neoverse N2 / Perseus	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	512 KB / 1 MB	System configuration
Neoverse V2 / Demeter	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	1 / 2 MB	System configuration
Neoverse V3	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	2 / 3 MB	System configuration
Neoverse N3	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	128 KB - 2 MB	System configuration
CSS N4	Licensee-defined; N4 targets 3 nm	IP, not a packaged SKU	up to 2 MB	System configuration

CPU IP comparison: implementation fields remain separate · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
Neoverse N1 / Ares	Implementation-defined	n/a	System integration	n/a	Armv8.2-A; NEON 128-bit
Neoverse E1	Implementation-defined	n/a	System integration	n/a	Armv8.2-A; NEON; SMT2
Neoverse V1 / Zeus	Implementation-defined	n/a	System integration	n/a	SVE: 2 x 256-bit; BF16; INT8
Neoverse N2 / Perseus	Implementation-defined	n/a	System integration	n/a	Armv9; SVE2 128-bit; BF16
Neoverse V2 / Demeter	Implementation-defined	n/a	System integration	n/a	Armv9; SVE2 128-bit; 4 vector pipes
Neoverse V3	Implementation-defined	n/a	System integration	n/a	Armv9.2-A; SVE2; CCA
Neoverse N3	Implementation-defined	n/a	System integration	n/a	Armv9.2-A; SVE2
CSS N4	Implementation-defined	n/a	System integration	n/a	Armv9; details n/d

Subsystem integration envelope

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Subsystem	Disclosure	Core envelope	Memory / I/O	Integration meaning
CSS N2	2023	Up to 64 N2	DDR5/LPDDR5; PCIe 5 / CXL	Prevalidated CPU and system design
CSS V3	2024	Up to 64 V3 per subsystem	up to 12 memory channels; 64 PCIe 5/CXL lanes	Chiplet-ready; CCA support
CSS N3	2024	Configurable N3 subsystem	CMN S3; CHI C2C	Cloud, networking, DPU and edge customization
CSS N4	2026-09	Up to 128 N4 per die	DDR5 / LPDDR6; PCIe 7 / CXL 4.0	3 nm target; configurable compute foundation

04 2015-2018: separating ISA, core and server

Before Neoverse, an Arm server could use licensed Cortex cores or a custom Arm-compatible microarchitecture. These are different business and engineering paths. Cortex-A57/A72-era designs provided a baseline, while Qualcomm Falkor, Cavium/Marvell ThunderX and Fujitsu's later A64FX demonstrate that an Arm ISA license does not imply a Neoverse core. That distinction remains essential for AmpereOne and NVIDIA Vera as well: their presence in the Arm ecosystem cannot be used as evidence about the internals of an Arm-designed core.

A server platform requires much more than an instruction decoder. Coherent I/O, interrupt virtualization, page-table maintenance, RAS, firmware boot conventions and a stable operating-system environment all affect deployment cost. Neoverse's 2018 branding and 2019 products made this infrastructure focus explicit. The lasting architectural advantage is the ability to assemble an application-specific balance of cores, caches, memory and accelerators, provided that integration and software costs remain manageable.

05 N1 and E1: cloud cores and throughput cores

N1 combines four-instruction decode with dispatch of up to eight internal micro-operations, a 128-entry commit queue, 64 KB instruction and data caches, and private L2. Its decoupled branch predictor runs ahead of instruction fetch; a 6K-entry main BTB helps large server code footprints. These widths describe different pipeline stages, not interchangeable measures of IPC. Direct attachment to CMN-600 lets designers grow the coherent system without imposing a fixed mobile cluster boundary. Server features include RAS, virtualization extensions and coherent instruction-cache behavior.

Hot Chips 2019 explains the architecture; ISSCC 2020 paper 8.3 supplies a concrete 3 GHz, 7 nm implementation anchor. The ISSCC summary emphasizes a short, operation-dependent pipeline and a low-latency memory hierarchy. Its PPA comparisons belong to that implementation and baseline, not every N1 licensee. E1 instead targets data-plane throughput with two hardware threads per core. It should not be treated as a low-clocked N1 or used to infer SMT in N-series and V-series products.

AWS Graviton2 and Ampere Altra turned N1 into commercial cloud and merchant-server products. Their usefulness comes from a complete platform and a large population of independent software threads, rather than from one universal Arm socket. Altra Max extends the density approach. Increasing cores while holding memory-channel count constant reduces the average external bandwidth budget per core; cache locality and workload consolidation determine whether that trade is productive.

06 V1 and N2: vector performance versus density

V1 creates a performance-oriented branch rather than replacing N1 in every role. Its two 256-bit SVE pipelines, BF16 support and INT8 matrix instructions increase CPU-side scientific and ML capability. Decoupled prediction, an instruction-side operation cache and a wider machine aim to keep the execution units busy. Vector-length-agnostic programming protects software from assuming one implementation width. AWS Graviton3 is the important cloud realization, but its chiplets and DDR5 channels are AWS design choices, not intrinsic V1 properties.

N2 takes Armv9 and 128-bit SVE2 into a more density-oriented core. The Hot Chips 2021 disclosure places the core together with CMN-700 and the wider system roadmap. The appropriate comparison with V1 is performance per area, per watt and per memory channel on the intended workload, not SVE width alone. Microsoft Cobalt 100 and networking implementations demonstrate how one core family can support different system balances. The CSS N2 offering in 2023 further reduces the amount of subsystem integration that a licensee must perform.

07 V2: width, memory concurrency and coherent hosts

V2's Hot Chips 2023 disclosure is unusually useful for reconstructing the whole core. It shows six-instruction decode, eight-wide dispatch and retirement, a 320-plus out-of-order window, six ALUs, two branch pipelines, and two load/store pipes plus one additional load pipe. Four 128-bit vector datapaths replace the V1 organization. The optional 2 MB private L2 retains a quoted ten-cycle load-to-use latency. Prefetch changes include indirect and page-table-walk handling, with lower priority for speculative prefetch traffic than demand requests. The design broadens useful concurrency while controlling memory interference.

V2 appears in AWS Graviton4, Google Axion and NVIDIA Grace. These systems illustrate three different optimizations: general cloud throughput, a vertically integrated cloud platform, and a coherent accelerator host. Grace's LPDDR5X and NVLink-C2C belong to NVIDIA's CPU and superchip integration. They do not make NVLink an Arm interconnect. Conversely, a cloud processor's DDR channels do not describe Grace. The common core provides software and implementation reuse; the system fabric and memory hierarchy determine the node-level behavior.

08 V3, N3 and CSS: moving the integration boundary

The 2024 generation makes a preintegrated subsystem a central product rather than a reference illustration. CSS V3 combines V3 cores, coherent fabric and platform IP, while CSS N3 serves a broader range of cloud, networking and edge configurations. V3 introduces Arm Confidential Computing Architecture support within the Neoverse family. CCA's Realm isolation model is separate from Morello's capability-based memory safety. Neither feature should be inferred merely because a chip implements some Armv9 instructions.

CSS trades some integration freedom for a verified baseline and a shorter path to silicon. The licensee still owns choices in packaging, memory population, clocks, firmware policy and commercial qualification. Microsoft Cobalt 200 uses CSS V3, yet its complete SoC has 132 active cores, 3 MB L2 per core and 192 MB system cache. Cobalt 200 entered customer preview in 2026. This is why a subsystem's core ceiling must not be mistaken for a package ceiling. AWS Graviton5 provides another V3-based cloud generation, reaching 192 cores per chip and becoming available through M9g/M9gd in June 2026.

09 AGI CPU: Arm becomes a silicon supplier

Announced in March 2026, AGI CPU turns the Neoverse platform into an Arm-designed server product. By the May shareholder update, commercial systems were available to order from named OEMs. Hot Chips 2026 adds the architecture context; the current product brief supplies SKU-level boundaries. The 136-core SP113012 has 2 MB L2 per core and 128 MB shared system cache, twelve DDR5-8800 channels, 96 PCIe 6 lanes and CXL 3.0 Type 3 support. Its maximum clock is 3.5 GHz; the 3.7 GHz headline belongs to the 64-core variant. All three listed variants have a 300 W base TDP setting.

The design emphasizes feeding many independent CPU tasks and attaching external accelerators. A theoretical 844.8 GB/s channel sum yields 6.21 GB/s per core at 136 cores and 13.2 GB/s at 64 cores. Those derived ratios explain the memory-per-core variant without implying sustained application bandwidth. The CPU remains a host and an inference-capable general processor. It is not an HBM tensor accelerator, and the 300 W CPU rating cannot be multiplied into a complete rack power estimate without memory, storage, NICs, fans and power-conversion losses.

Arm merchant CPU reference SKUs - 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Codename	GA	Status	Core µarch	Cores / threads
AGI CPU 136C	SP113012	2026	Shipping / available	Neoverse V3	136 / 136
AGI CPU 128C	SP113012S	2026	Shipping / available	Neoverse V3	128 / 128
AGI CPU 64C	SP113012A	2026	Shipping / available	Neoverse V3	64 / 64

Arm merchant CPU reference SKUs - 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Process per die	Dies / packaging	L2 per core	L3 / domain
AGI CPU 136C	3 nm	Arm-designed server SoC	2 MB	128 MB SLC
AGI CPU 128C	3 nm	Arm-designed server SoC	2 MB	128 MB SLC
AGI CPU 64C	3 nm	Arm-designed server SoC	2 MB	128 MB SLC

Arm merchant CPU reference SKUs - 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Generation / reference SKU	Memory: channels; rate; GB/s	Sockets / links	PCIe / CXL	TDP	Vector / matrix ISA
AGI CPU 136C	DDR5; 12; 8800 MT/s; 844.8 GB/s	2S; link rate n/d	96 PCIe 6; CXL 3.0 Type 3	300 W	2 x 128-bit SVE2; BF16; INT8 MMLA
AGI CPU 128C	DDR5; 12; 8800 MT/s; 844.8 GB/s	2S; link rate n/d	96 PCIe 6; CXL 3.0 Type 3	300 W	2 x 128-bit SVE2; BF16; INT8 MMLA
AGI CPU 64C	DDR5; 12; 8800 MT/s; 844.8 GB/s	2S; link rate n/d	96 PCIe 6; CXL 3.0 Type 3	300 W	2 x 128-bit SVE2; BF16; INT8 MMLA

10 CSS N4 and the current roadmap

September 2026 CSS N4 is a new IP/subsystem generation, not the core inside the first AGI CPU. Arm lists up to 128 N4 cores per die, private L2 configurable to 2 MB, a 3 nm implementation target, DDR5 or LPDDR6, and PCIe 7/CXL 4.0 I/O. These are subsystem capabilities; a licensee need not instantiate every maximum together. The architectural direction is a wider deployment envelope, including CPU hosts and infrastructure processors whose memory and I/O requirements differ.

A public IP announcement does not establish final commercial silicon clocks, socket TDP, DDR qualification or deployment dates. The reviewed evidence does not provide a complete N4 core queue-and-port disclosure comparable with the V2 Hot Chips talk. The report therefore preserves the new generation and its verified interfaces without filling those fields from client-core analogies. Mobile CSS and automotive V3AE derivatives show reuse, but their safety, graphics and power domains remain distinct from the cloud subsystem.

11 CMN, CHI and the infrastructure offload boundary

CMN-600, CMN-700 and CMN S3 make coherence a configurable system resource. Request nodes inject CPU and I/O transactions; home nodes coordinate address ownership, directories and system cache; memory-facing nodes connect to controllers. CHI specifies transaction and coherence behavior, while the mesh implements routing and buffering. Consequently, a CHI data-channel width is not a SerDes lane rate and a mesh peak is not DRAM bandwidth. Chiplet crossings add a physical-link and latency problem beyond on-die coherence.

Arm cores inside DPUs and IPU provide a programmable control or service processor. Packet pipelines, RDMA engines, crypto, local memory and ports are supplied by the chip designer. Arm's participation in a DPU therefore does not make that device an Arm-branded NIC. CSS N3 and the Alphawave collaboration show how a compute chiplet can be combined with a networking or acceleration die. The useful unit of comparison is the whole device: host interface, offload engines, transport semantics and software isolation.

Interconnect ownership and semantics · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Year / generation	Lane rate	Lanes / link	GB/s per link per direction
CMN-600 / CHI	N1	n/a	n/a	n/d
CMN-700	N2 / V2	n/a	n/a	n/d
CMN S3 / CHI C2C	V3 / N3	n/d	n/d	n/d
AGI CPU PCIe 6 / CXL 3	2026	64 GT/s	up to x16	~128 raw x16

Interconnect ownership and semantics · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Interconnect	Links / device	Topology	Domain limit	Coherence / semantics
CMN-600 / CHI	n/d	Coherent on-die mesh	Implementation-defined	Cache coherence; system cache
CMN-700	n/d	Coherent mesh; multi-chip integration	n/d	CHI
CMN S3 / CHI C2C	n/d	Mesh and chiplet links	n/d	Coherent subsystem integration
AGI CPU PCIe 6 / CXL 3	96 lanes total	Host I/O fabric	n/d	PCIe I/O; CXL Type 3 memory

Network product boundary · 1/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	GA / milestone	Status	Class	Ports × speed	SerDes
No Arm-branded AI NIC established	n/a	n/a	IP supplier / CPU supplier	n/a	n/a

Network product boundary · 2/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Host interface	Programmable engine	Embedded CPU	On-card memory
No Arm-branded AI NIC established	n/a	n/a	Neoverse licensed to device vendors	n/a

Network product boundary · 3/3

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Transport / RDMA	Congestion control	Key offloads	Power
No Arm-branded AI NIC established	Licensee-specific	n/a	CPU control; packet engines separate	n/a

12 GPU and NPU IP: complementary edge compute

Mali evolves through Midgard, Bifrost, Valhall and later graphics generations; Immortalis adds a premium graphics branch. These designs target integrated graphics and shared-memory SoCs. Their presence does not establish a merchant HBM GPU for distributed training. The 2026 Mali G2-Ultra NX adds dedicated neural acceleration within the mobile graphics platform, alongside the C2 CPU cluster and SME2. This is important convergence at the edge, while its bandwidth, power and software environment remain different from a datacenter accelerator card.

Ethos-N77/N78 address application-processor and edge inference integration; Ethos-U55/U65/U85 address constrained embedded systems. U65 extends the U55 approach into Cortex-A, Cortex-R and Neoverse systems. U85 expands to 128-2048 MAC units and supports transformer-oriented operators. At 1 GHz, 2048 MACs correspond to 4.096 TOPS when one multiply-add counts as two integer operations; this is a configuration calculation, not BF16 throughput. Compiler partitioning, SRAM tiling and unsupported-operator fallback determine practical model coverage.

Complementary accelerator IP in the common schema · 1/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Architecture	GA / milestone	Status	Process	Dies / packaging
Ethos-N78	Ethos-N	2020 IP	Licensable IP	n/a	Integrated IP
Ethos-U85	Ethos-U	2024 IP	Licensable IP	n/a	Integrated IP
Mali G2-Ultra NX	Mali	2026 IP	Announced; GA n/d	n/a	Mobile GPU IP

Complementary accelerator IP in the common schema · 2/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Compute units	Dense BF16	Dense FP8	Dense FP6 / FP4	FP64 vector / matrix
Ethos-N78	configurable	n/d	n/d	n/d	n/a
Ethos-U85	128-2048 MACs	n/d	n/d	n/d	n/a
Mali G2-Ultra NX	n/d	n/d	n/d	n/d	n/d

Complementary accelerator IP in the common schema · 3/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Memory: type; stacks; GB; TB/s	LLC / SRAM	Scale-up: links; GB/s per direction; domain
Ethos-N78	HBM n/a; SoC memory	n/d	n/a
Ethos-U85	HBM n/a; system SRAM/DRAM	n/d	n/a
Mali G2-Ultra NX	HBM n/a; shared SoC memory	n/d	n/a

Complementary accelerator IP in the common schema · 4/4

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Product	Host link	Power / cooling	Form factor
Ethos-N78	SoC interconnect	n/a	IP
Ethos-U85	SoC interconnect	n/a	IP
Mali G2-Ultra NX	SoC interconnect	n/a	IP

13 Software, security and research evidence

Arm's software advantage comes from a common AArch64 ecosystem combined with implementation-specific optimization. GCC/LLVM targets, Linux scheduling and NUMA placement, optimized math kernels, Compute Library and profiling tools must match the actual core and memory system. SVE-length-agnostic code still needs sensible blocking and cache reuse. CSS reference software supplies firmware and platform models; it does not automatically qualify a production server. In AI hosting, orchestration latency, tokenizer performance and communication progress can matter even when all large matrix operations run elsewhere.

The academic record adds mechanisms and evaluation methods rather than a complete product roadmap. Arm-associated MISB at ISCA 2019 and temporal prefetching at MICRO 2019 study the storage and traffic cost of prediction metadata. Triangel at ISCA 2024 studies timeliness and accuracy; HPCA 2025 bandwidth-partitioning work evaluates resource interference; ASPLOS 2025 hierarchical prefetching addresses large instruction footprints. These help explain what designers are optimizing. They do not prove that a named production core implements the proposed mechanism. Morello is different: its documented N1-derived evaluation platform is an explicit research-to-hardware link, but remains a capability-security prototype rather than a standard Neoverse feature.

Software milestones

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Release / component	Date	Hardware enabled	Key features
AArch64 Linux / GCC / LLVM	2015-2026	Cortex, Neoverse and custom cores	Portable binaries; core-specific scheduling and ISA dispatch
Arm Compute Library	ongoing	Arm CPU / Mali	Optimized AI operators; feature-dependent paths
Neoverse reference software	2024 release	RD-V3 and related platforms	Firmware and fixed virtual platform integration

14 Systems and quantitative balance

Representative system boundaries · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Year / milestone	CPU / accelerator count	Scale-up domain
Licensee cloud server	2019-2026	Graviton / Axion / Cobalt	Provider-specific
AGI CPU 10U dual-node reference	2026	2 x AGI CPU	Two independent nodes
AGI CPU 2U2P reference	2026	2 x AGI CPU	2-socket coherent server

Representative system boundaries · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Platform	Scale-out per accelerator	Rack power	Cooling
Licensee cloud server	n/d	n/d	n/d
AGI CPU 10U dual-node reference	n/d	36 kW example; 8160 cores	Air
AGI CPU 2U2P reference	n/d	n/d	Air

Derived bandwidth and cache ratios

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Configuration	DRAM peak GB/s	GB/s/core	SLC MB/core	Meaning
AGI 136C	844.8	6.21	0.94	Equal-share ratio; not a reservation
AGI 128C	844.8	6.60	1.00	Same interface, fewer active cores
AGI 64C	844.8	13.20	2.00	Memory and cache capacity per core increase
Graviton4 96C	537.6	5.60	n/d	12 x DDR5-5600; AWS implementation

HBM bytes per dense BF16/FP8 FLOP and accelerator scale-up/HBM ratios are n/a for the Arm-branded CPU/IP portfolio. Assigning a partner GPU's HBM to Arm would corrupt a cross-vendor comparison. Likewise, no universal tokens/W number follows from core count or an IP efficiency projection. An end-to-end measurement needs the actual model, precision, latency target, memory configuration and system power boundary.

15 Conference-to-product index

Verified disclosure anchors · 1/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Year	Paper / talk / disclosure	Presenter / authors
Hot Chips	2019	Neoverse N1 Cloud-to-Edge Infrastructure SoCs	Arm
ISSCC	2020	A 3GHz Arm Neoverse N1 CPU in 7nm FinFET for Infrastructure Applications	R. Christy et al.
Hot Chips	2021	Arm Neoverse N2: 2nd generation infrastructure CPUs and system IPs	A. Pellegrini
Hot Chips	2023	Arm Neoverse V2 Platform	Magnus Bruce
Hot Chips	2023	Neoverse Compute Subsystems launch	Arm
Hot Chips	2026	Arm AGI CPU architecture disclosure	Arm
ISCA	2019 / 2024	MISB; Triangel	Arm / academic researchers
MICRO	2019	Temporal Prefetching Without the Off-Chip Metadata	Wu et al.
HPCA	2025	Criticality-Aware Instruction-Centric Bandwidth Partitioning	Paper authors
ASPLOS	2025	Hierarchical Prefetching	Zhang et al.
Microsoft Ignite / Build	2025 / 2026	Cobalt 200 disclosure and preview	Microsoft
Google Cloud Next	2024	Google Axion introduction	Google
AWS re:Invent / EC2 launches	2023-2026	Graviton4 / Graviton5	AWS
Arm Everywhere	2026	AGI CPU and CSS N4	Arm

Verified disclosure anchors · 2/2

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Venue	Product	Evidence type	Source
Hot Chips	N1	Architecture	P01
ISSCC	N1	Circuit / implementation	C01
Hot Chips	N2 / CMN-700	Architecture	C04
Hot Chips	V2	Architecture	C02
Hot Chips	CSS N2	Subsystem	E03
Hot Chips	AGI CPU	Architecture / system	C03
ISCA	Related memory research	Related research	R06 R03
MICRO	Related memory research	Related research	R05
HPCA	Resource partitioning	Related research	R04
ASPLOS	Instruction supply	Related research	R07
Microsoft Ignite / Build	CSS V3 implementation	Partner deployment	E07 E18 E20
Google Cloud Next	V2 implementation	Partner deployment	E17
AWS re:Invent / EC2 launches	V2 / V3 implementations	Partner deployment	E16 E19
Arm Everywhere	Merchant CPU / IP	Product launch	E05 E06

DAC, MWC, Computex, OCP, GTC, SC/ISC, VLSI and ISSCC/JSSC are complementary channels: design methodology, telecom deployments, ecosystem launches, racks, coherent GPU hosts, HPC systems and implementation technology. They are included when a specific disclosure is tied to a covered generation. The index is a verified selection, not a claim that Arm presented every product at every venue. The N1 ISSCC entry uses the conference's published technical summary; it does not imply access to the complete paper.

16 Codename and ownership crosswalk

Names that must not be conflated

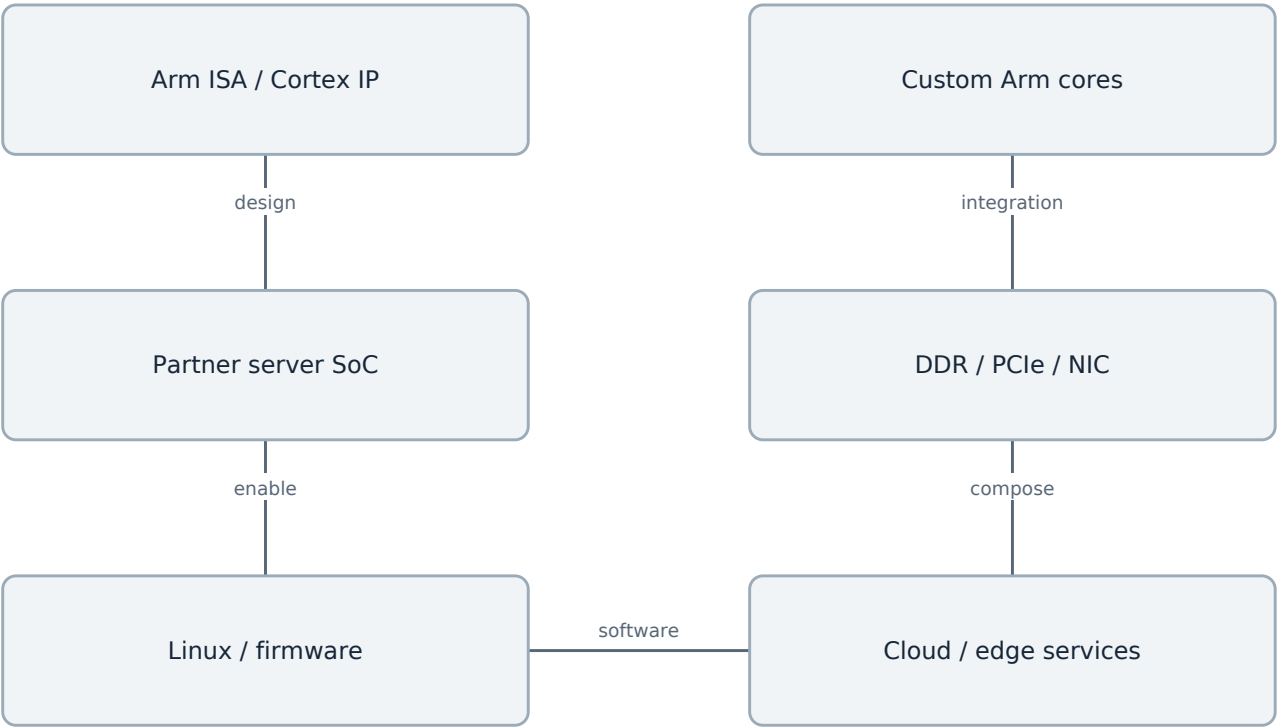
n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Name	Meaning	Owner / boundary
Ares / N1	Core IP generation	Arm
Zeus / V1	Vector-oriented core IP	Arm
Perseus / N2	Density-oriented Armv9 core	Arm
Demeter / V2	Core used by Grace, Axion and Graviton4	Arm core; partner SoCs
CSS	Preintegrated compute subsystem	Not a complete merchant chip
AGI CPU	Arm-designed server silicon	V3-based; not CSS N4
Falkor / Oryon / AmpereOne / Vera	Custom Arm-compatible cores or products	Licensee designs; not Neoverse aliases

17 Integrated stack across four eras

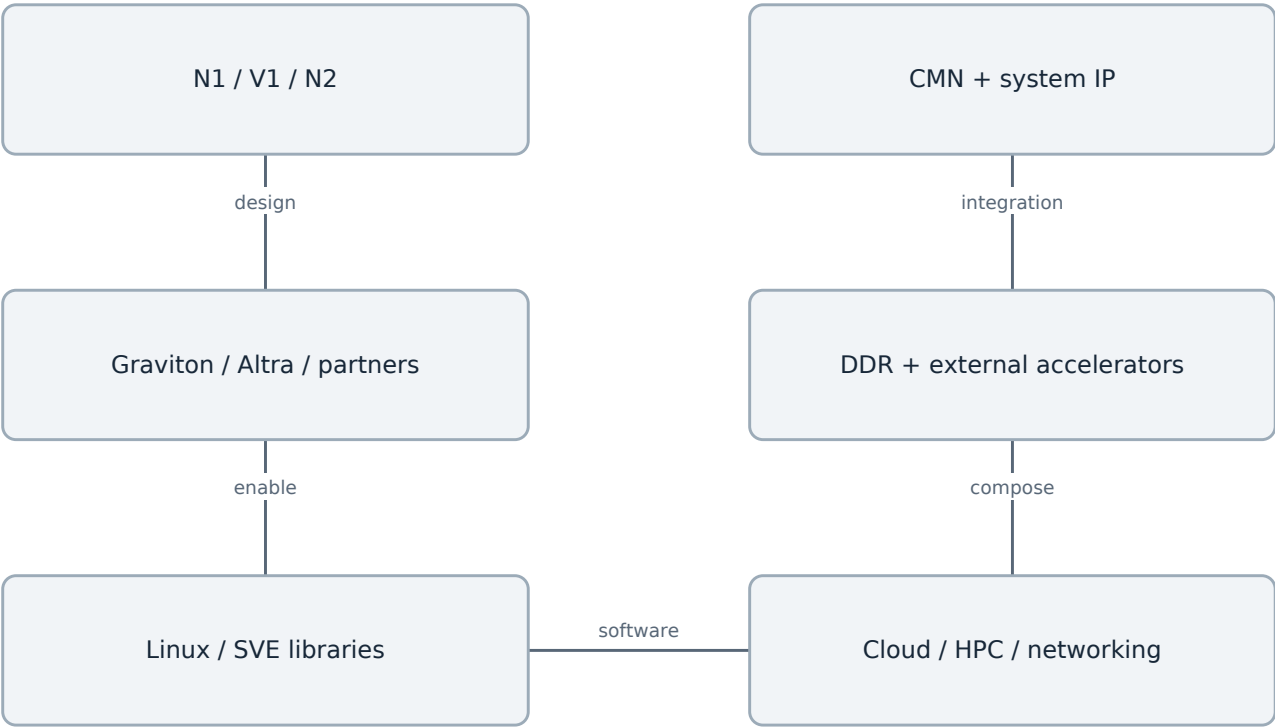
These figures show ownership and composition, not an assertion of a common physical fabric across all licensees. Connections indicate integration relationships. Exact bandwidth remains implementation-specific.

2015-2021: ISA and partner integration



Conceptual ownership and integration map; partner GPU and network bandwidth is not attributed to Arm.

2019-2022: Neoverse platforms



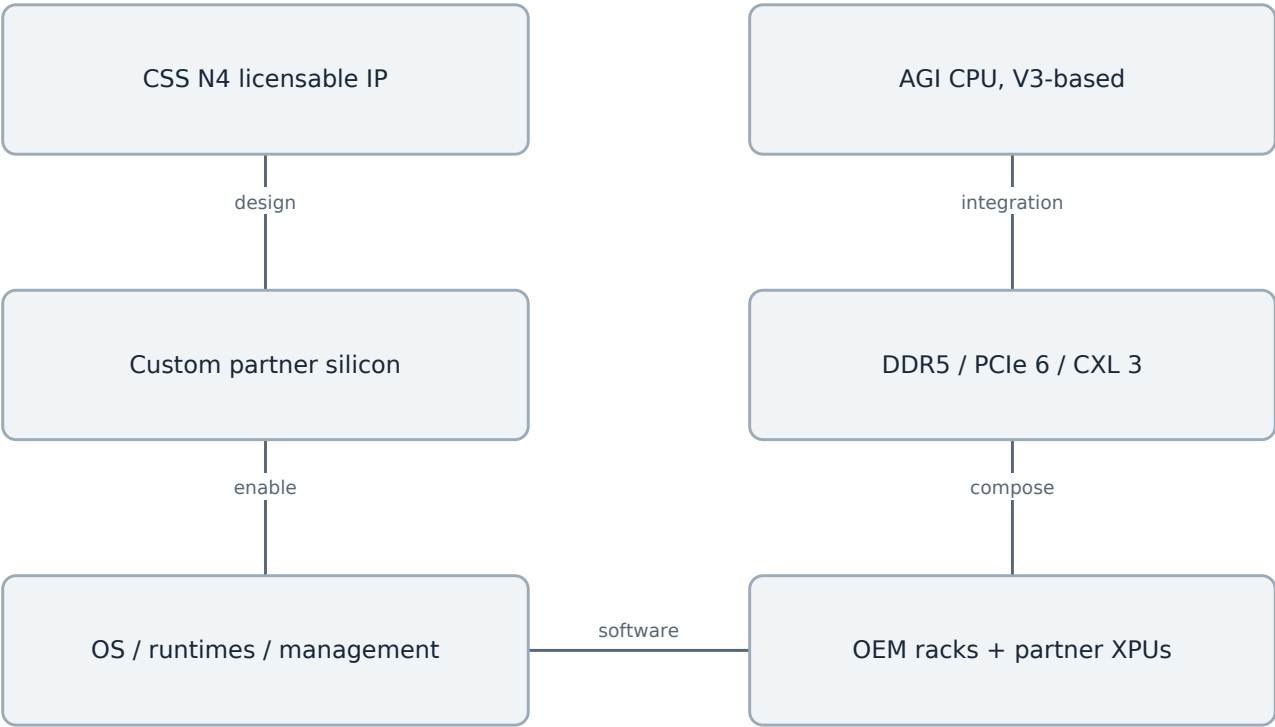
Conceptual ownership and integration map; partner GPU and network bandwidth is not attributed to Arm.

2023-2025: CSS and heterogeneous hosts



Conceptual ownership and integration map; partner GPU and network bandwidth is not attributed to Arm.

2026: IP and merchant silicon coexist



Conceptual ownership and integration map; partner GPU and network bandwidth is not attributed to Arm.

Bandwidth hierarchy by era

n/d: not established by reviewed sources; n/a: not applicable. See Section 02 for numerical conventions and the source map at the end.

Era	On-die / die-to-die	Memory	Host attach	Scale-up / scale-out
2015-2018	Licensee fabric	Licensee DDR	PCIe; n/d	Licensee network; n/d
2019-2022	CMN-600/700; n/d	Partner-configured DDR; n/d	PCIe / CCIX; n/d	No universal Arm link
2023-2025	CMN / CHI C2C; n/d	Graviton4: 537.6 GB/s peak	Grace C2C is NVIDIA-owned	Partner-specific
2026	CMN; D2D n/d	AGI: 844.8 GB/s derived peak	PCIe 6 x16 ~128 GB/s raw	External XPU/NIC; n/d

18 Engineering conclusions

Arm's decade is best understood as the gradual productization of a configurable system. Core efficiency matters, but the larger gains come from pairing the right core with the right memory, coherence and deployment model. CSS reduces integration cost without removing implementation responsibility. Merchant AGI silicon widens access to the platform while introducing a second relationship with existing licensees. For AI infrastructure, the enduring CPU requirements are predictable service latency, memory capacity and bandwidth, accelerator I/O, isolation and mature software. Neither an Arm ISA label nor a maximum core count settles those questions.

Source map

- 01 Scope and executive synthesis - C02, E01, E03, E04, E05, E06, E15, P01, P02, P06
- 02 Units, ownership and availability - C02, E08, P02, P04, P07, P08
- 03 Neoverse and CSS generation atlas - C02, E01, E02, E03, E04, E06, E09, P01, P02, P03, P04, P06, P08, P09, P10
- 04 2015-2018: separating ISA, core and server - C03, E01, E08, P01, P08
- 05 N1 and E1: cloud cores and throughput cores - C01, E01, E08, P01, P08
- 06 V1 and N2: vector performance versus density - C04, E02, E03, E08, P02, P03
- 07 V2: width, memory concurrency and coherent hosts - C02, E08, E16, E17, P15
- 08 V3, N3 and CSS: moving the integration boundary - E04, E07, E18, E19, E20, P04, P08, P09, R01
- 09 AGI CPU: Arm becomes a silicon supplier - C03, E05, E13, E14, E15, P07
- 10 CSS N4 and the current roadmap - E06, E10, P06, P10
- 11 CMN, CHI and the infrastructure offload boundary - C02, C04, E09, P01, P07, P08, P09, P14
- 12 GPU and NPU IP: complementary edge compute - E10, E11, E12, P11, P12
- 13 Software, security and research evidence - P08, P13, P14, R01, R02, R03, R04, R05, R06, R07
- 14 Systems and quantitative balance - E08, E13, E14, E16, E17, E18, E19, P07, P11
- 15 Conference-to-product index - C01, C02, C03, C04, E03, E05, E06, E07, E09, E13, E16, E17, E18, E19, E20, P01, P15, R03, R04, R05, R06, R07
- 16 Codename and ownership crosswalk - C02, C03, E01, E03, E06, E08, P01, P02, P07, P08
- 17 Integrated stack across four eras - C04, E03, E06, E08, E09, E16, P01, P07, P15
- 18 Engineering conclusions - E03, E06, P07, P13

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